

Regularizing Choi Covariance Contractions

Let the total circuit operator at time step t be $\hat{C}_t$.

$$\text{i.e. } |\Psi_t\rangle = \frac{\hat{C}_t |\Psi_0\rangle}{\|\hat{C}_t |\Psi_0\rangle\|}$$

The operator $\hat{C}_t$ evolves as $\hat{C}_{t+1} = \hat{K}_{t+1} \hat{C}_t$, Where $\hat{K}_{t+1}$ can be

$$\hat{K}_{t+1} \in \{\hat{N}_-, \hat{X}_-^\dagger (1 - \hat{N}_-), (1 - \hat{N}_+), \hat{X}_+ \hat{N}_+\}$$

Where the $\pm$ subscripts denote band index. The Choi covariance representation is determined by a single particle projector

$$P_\pm = \vec{X}_\pm \vec{X}_\pm^\dagger, \text{ Where } \vec{X}_\pm^\dagger = \sum_i (\vec{X}_\pm)_i \hat{c}_i^\dagger$$

In Tensor network form, we have

$$\begin{array}{c} L \\ \text{---} \end{array} \textcircled{\Sigma_{t+1}} \begin{array}{c} R \\ \text{---} \end{array} = \begin{array}{c} L \\ \text{---} \end{array} \textcircled{I_t} \text{---} \textcircled{K_{t+1}} \begin{array}{c} R \\ \text{---} \end{array}$$

$$\begin{array}{c} L \\ \text{---} \end{array} \textcircled{\Sigma_t} \begin{array}{c} R \\ \text{---} \end{array} = \begin{pmatrix} \Sigma_{LL} & \Sigma_{LR} \\ \Sigma_{LR}^\dagger & \Sigma_{RR} \end{pmatrix} ; \quad \begin{array}{c} L \\ \text{---} \end{array} \textcircled{K_{t+1}} \begin{array}{c} R \\ \text{---} \end{array} = \begin{pmatrix} K_{LL} & K_{LR} \\ K_{LR}^\dagger & K_{RR} \end{pmatrix}$$

For the four kind of operations, we have the four different Choi covariances

$$1) \hat{K}_{t+1} = \hat{N}_- \Rightarrow \begin{pmatrix} K_{LL} & K_{LR} \\ K_{LR}^\dagger & K_{RR} \end{pmatrix} = \begin{pmatrix} -P_- & 1-P_- \\ 1-P_- & P_- \end{pmatrix}$$

$$2) \hat{K}_{t+1} = \hat{X}_-^\dagger (1 - \hat{N}_-) \Rightarrow \begin{pmatrix} K_{LL} & K_{LR} \\ K_{LR}^\dagger & K_{RR} \end{pmatrix} = \begin{pmatrix} P_- & 1-P_- \\ 1-P_- & P_- \end{pmatrix}$$

$$3) \hat{K}_{t+1} = 1 - \hat{N}_+ \Rightarrow \begin{pmatrix} K_{LL} & K_{LR} \\ K_{LR}^\dagger & K_{RR} \end{pmatrix} = \begin{pmatrix} P_+ & 1-P_+ \\ 1-P_+ & -P_+ \end{pmatrix}$$

$$4) \hat{K}_{t+1} = \hat{X}_+ \hat{N}_+ \Rightarrow \begin{pmatrix} K_{LL} & K_{LR} \\ K_{LR}^\dagger & K_{RR} \end{pmatrix} = \begin{pmatrix} -P_+ & 1-P_+ \\ 1-P_+ & -P_+ \end{pmatrix}$$

We can handle all 4 possibilities at once using two independent sign parameters and dropping band indices

$$\begin{pmatrix} K_{LL} & K_{LR} \\ K_{LR}^\dagger & K_{RR} \end{pmatrix} = \begin{pmatrix} \eta_1 P & 1-P \\ 1-P & \eta_2 P \end{pmatrix}, \quad \eta_1, \eta_2 = \pm$$

Now apply the contraction formula (denote contraction by $\circ$)

$$\begin{array}{c} L \\ \hline \end{array} \text{---} \text{---} \text{---} \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \begin{array}{c} R \\ \hline \end{array} = \begin{array}{c} L \\ \hline \end{array} \text{---} \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \text{---} \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \begin{array}{c} R \\ \hline \end{array} = \begin{pmatrix} \Sigma_{LL} & \Sigma_{LR} \\ \Sigma_{LR}^\dagger & \Sigma_{RR} \end{pmatrix} \circ \begin{pmatrix} K_{LL} & K_{LR} \\ K_{LR}^\dagger & K_{RR} \end{pmatrix}$$

$$= \begin{pmatrix} \Sigma_{LL} & \\ & K_{RR} \end{pmatrix} - \begin{pmatrix} \Sigma_{LR} & \\ & K_{LR}^\dagger \end{pmatrix} \begin{pmatrix} \Sigma_{RR} & -1 \\ -1 & K_{LL} \end{pmatrix}^{-1} \begin{pmatrix} \Sigma_{LR}^\dagger & \\ & K_{LR} \end{pmatrix}$$

$$\begin{pmatrix} \Sigma_{RR} & -1 \\ -1 & K_{LL} \end{pmatrix}^{-1} = \begin{pmatrix} (\Sigma_{RR} - K_{LL}^{-1})^{-1} & (\Sigma_{RR} - K_{LL}^{-1})^{-1} K_{LL}^{-1} \\ K_{LL}^{-1} (\Sigma_{RR} - K_{LL}^{-1})^{-1} & K_{LL}^{-1} + K_{LL}^{-1} (\Sigma_{RR} - K_{LL}^{-1})^{-1} K_{LL}^{-1} \end{pmatrix}$$

Now suppose $K_{LL} = \eta_1 P$ is rank-1 i.e. non-invertible

$$\begin{aligned} \text{Introduce a regulator via } K_{LL} &\rightarrow K_{LL}^\varepsilon = \eta_1 P + \varepsilon(1-P) \sim \begin{pmatrix} \eta_1 & \\ & \varepsilon 1 \end{pmatrix} \\ \Rightarrow (K_{LL}^\varepsilon)^{-1} &= \eta_1 P + \frac{1}{\varepsilon}(1-P) \sim \begin{pmatrix} \eta_1 & \\ & \frac{1}{\varepsilon} 1 \end{pmatrix} \end{aligned}$$

Now work in the diagonal basis of P and parametrize

$$\Sigma_{RR} = \begin{pmatrix} a & \vec{b}^\dagger \\ \vec{b} & c \end{pmatrix} \quad \text{Then } (\Sigma_{RR} - K_{LL}^{\varepsilon^{-1}})^{-1} = \begin{pmatrix} a - \eta_1 & \vec{b}^\dagger \\ \vec{b} & c - \frac{1}{\varepsilon} 1 \end{pmatrix}^{-1}.$$

Using $(c - \frac{1}{\varepsilon} 1)^{-1} = -\varepsilon(1 - \varepsilon c)^{-1}$, we have

$$\text{Then } \begin{pmatrix} a - \eta_1 & \vec{b}^\dagger \\ \vec{b} & c - \frac{1}{\varepsilon} 1 \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} (a - \eta_1 + \varepsilon \vec{b}^\dagger (1 - \varepsilon c)^{-1} \vec{b})^{-1} & (a - \eta_1 + \varepsilon \vec{b}^\dagger (1 - \varepsilon c)^{-1} \vec{b})^{-1} \vec{b}^\dagger \varepsilon (1 - \varepsilon c)^{-1} \\ \varepsilon (1 - \varepsilon c)^{-1} \vec{b} (a - \eta_1 + \varepsilon \vec{b}^\dagger (1 - \varepsilon c)^{-1} \vec{b})^{-1} & -\varepsilon (1 - \varepsilon c)^{-1} + \varepsilon (1 - \varepsilon c)^{-1} \vec{b} (a - \eta_1 + \varepsilon \vec{b}^\dagger (1 - \varepsilon c)^{-1} \vec{b})^{-1} \vec{b}^\dagger \varepsilon (1 - \varepsilon c)^{-1} \end{pmatrix}$$

Expand everything to $\mathcal{O}(\varepsilon^2)$

$$\begin{aligned}
 1) (a - \eta_l + \varepsilon \vec{b}^\dagger (1 - \varepsilon C)^{-1} \vec{b})^{-1} &= (a - \eta_l + \varepsilon \vec{b}^\dagger (1 + \varepsilon C) \vec{b} + \mathcal{O}(\varepsilon^3))^{-1} \\
 &= \frac{1}{a - \eta_l} \left[1 + \frac{\varepsilon}{a - \eta_l} \vec{b}^\dagger (1 + \varepsilon C) \vec{b} + \mathcal{O}(\varepsilon^3) \right] = \frac{1}{a - \eta_l} \left[1 - \frac{\varepsilon}{a - \eta_l} \vec{b}^\dagger (1 + \varepsilon C) \vec{b} + \frac{\varepsilon^2}{(a - \eta_l)^2} (\vec{b}^\dagger (1 + \varepsilon C) \vec{b})^2 + \mathcal{O}(\varepsilon^3) \right] \\
 &= \frac{1}{a - \eta_l} \left[1 - \frac{\varepsilon}{a - \eta_l} \vec{b}^\dagger \vec{b} + \frac{\varepsilon^2}{a - \eta_l} (\vec{b}^\dagger C \vec{b} + \frac{1}{a - \eta_l} (\vec{b}^\dagger \vec{b})^2) \right] + \mathcal{O}(\varepsilon^3)
 \end{aligned}$$

$$\begin{aligned}
 2) (a - \eta_l + \varepsilon \vec{b}^\dagger (1 - \varepsilon C)^{-1} \vec{b})^{-1} \vec{b}^\dagger \varepsilon (1 - \varepsilon C)^{-1} \\
 &= \frac{1}{a - \eta_l} \left[1 - \frac{\varepsilon}{a - \eta_l} \vec{b}^\dagger \vec{b} + \frac{\varepsilon^2}{a - \eta_l} (\vec{b}^\dagger C \vec{b} + \frac{1}{a - \eta_l} (\vec{b}^\dagger \vec{b})^2) \right] \vec{b}^\dagger \varepsilon (1 + \varepsilon C) + \mathcal{O}(\varepsilon^3) \\
 &= \frac{1}{a - \eta_l} \left[\varepsilon 1 - \frac{\varepsilon^2}{a - \eta_l} \vec{b}^\dagger \vec{b} \right] \vec{b}^\dagger + \mathcal{O}(\varepsilon^3)
 \end{aligned}$$

$$\begin{aligned}
 3) -\varepsilon (1 - \varepsilon C)^{-1} + \varepsilon (1 - \varepsilon C)^{-1} \vec{b} (a - \eta_l + \varepsilon \vec{b}^\dagger (1 - \varepsilon C)^{-1} \vec{b})^{-1} \vec{b}^\dagger \varepsilon (1 - \varepsilon C)^{-1} \\
 &= -\varepsilon 1 - \varepsilon^2 C + \varepsilon^2 \frac{\vec{b} \vec{b}^\dagger}{a - \eta_l} = -\varepsilon 1 + \varepsilon^2 \left(\frac{1}{a - \eta_l} \vec{b} \vec{b}^\dagger - C \right) + \mathcal{O}(\varepsilon^3)
 \end{aligned}$$

$$\Rightarrow (\mathcal{I}_{RR} - K_{LL}^{\varepsilon-1})^{-1} = \begin{pmatrix} a - \eta_l & \vec{b}^\dagger \\ \vec{b} & C - \frac{1}{\varepsilon} 1 \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} \frac{1}{a - \eta_l} \left[1 - \frac{\varepsilon}{a - \eta_l} \vec{b}^\dagger \vec{b} + \frac{\varepsilon^2}{a - \eta_l} (\vec{b}^\dagger C \vec{b} + \frac{1}{a - \eta_l} (\vec{b}^\dagger \vec{b})^2) \right] & \frac{1}{a - \eta_l} \left[\varepsilon 1 - \frac{\varepsilon^2}{a - \eta_l} \vec{b}^\dagger \vec{b} \right] \vec{b}^\dagger \\ \vec{b} \frac{1}{a - \eta_l} \left[\varepsilon 1 - \frac{\varepsilon^2}{a - \eta_l} \vec{b}^\dagger \vec{b} \right] & -\varepsilon 1 + \varepsilon^2 \left(\frac{1}{a - \eta_l} \vec{b} \vec{b}^\dagger - C \right) \end{pmatrix} + \mathcal{O}(\varepsilon^3)$$

Now consider each block of $\begin{pmatrix} (\mathcal{I}_{RR} - K_{LL}^{-1})^{-1} & (\mathcal{I}_{RR} - K_{LL}^{-1})^{-1} K_{LL}^{-1} \\ K_{LL}^{-1} (\mathcal{I}_{RR} - K_{LL}^{-1})^{-1} & K_{LL}^{-1} + K_{LL}^{-1} (\mathcal{I}_{RR} - K_{LL}^{-1})^{-1} K_{LL}^{-1} \end{pmatrix}$

and take $\varepsilon \rightarrow 0$

$$1) (\Sigma_{RR} - K_{LL}^{-1})^{-1} \xrightarrow{\varepsilon \rightarrow 0} \frac{1}{\alpha - \eta_1} P$$

$$2) (\Sigma_{RR} - K_{LL}^{-1})^{-1} K_{LL}^{-1} =$$

$$\begin{pmatrix} \frac{1}{\alpha - \eta_1} \left[1 - \frac{\varepsilon}{\alpha - \eta_1} \vec{b}^T \vec{b} + \frac{\varepsilon^2}{\alpha - \eta_1} (\vec{b}^T c \vec{b} + \frac{1}{\alpha - \eta_1} (\vec{b}^T \vec{b})^2) \right] & \frac{1}{\alpha - \eta_1} \left[\varepsilon 1 - \frac{\varepsilon^2}{\alpha - \eta_1} \vec{b}^T \vec{b} \right] \vec{b}^T \\ \vec{b} \frac{1}{\alpha - \eta_1} \left[\varepsilon 1 - \frac{\varepsilon^2}{\alpha - \eta_1} \vec{b}^T \vec{b} \right] & -\varepsilon 1 + \varepsilon^2 \left(\frac{1}{\alpha - \eta_1} \vec{b} \vec{b}^T - c \right) \end{pmatrix} \begin{pmatrix} \eta_1 \\ \frac{1}{\varepsilon} 1 \end{pmatrix} \xrightarrow{\varepsilon \rightarrow 0} \begin{pmatrix} \frac{\eta_1}{\alpha - \eta_1} & \frac{1}{\alpha - \eta_1} \vec{b}^T \\ 0 & -1 \end{pmatrix}$$

$$3) K_{LL}^{-1} (\Sigma_{RR} - K_{LL}^{-1})^{-1} \xrightarrow{\varepsilon \rightarrow 0} \begin{pmatrix} \frac{\eta_1}{\alpha - \eta_1} & 0 \\ \frac{1}{\alpha - \eta_1} \vec{b} & -1 \end{pmatrix}$$

$$4) K_{LL}^{-1} + K_{LL}^{-1} (\Sigma_{RR} - K_{LL}^{-1})^{-1} K_{LL}^{-1} =$$

$$\begin{pmatrix} \eta_1 \\ \frac{1}{\varepsilon} 1 \end{pmatrix} + \begin{pmatrix} \eta_1 \\ \frac{1}{\varepsilon} 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\alpha - \eta_1} \left[1 - \frac{\varepsilon}{\alpha - \eta_1} \vec{b}^T \vec{b} + \frac{\varepsilon^2}{\alpha - \eta_1} (\vec{b}^T c \vec{b} + \frac{1}{\alpha - \eta_1} (\vec{b}^T \vec{b})^2) \right] & \frac{1}{\alpha - \eta_1} \left[\varepsilon 1 - \frac{\varepsilon^2}{\alpha - \eta_1} \vec{b}^T \vec{b} \right] \vec{b}^T \\ \vec{b} \frac{1}{\alpha - \eta_1} \left[\varepsilon 1 - \frac{\varepsilon^2}{\alpha - \eta_1} \vec{b}^T \vec{b} \right] & -\varepsilon 1 + \varepsilon^2 \left(\frac{1}{\alpha - \eta_1} \vec{b} \vec{b}^T - c \right) \end{pmatrix} \begin{pmatrix} \eta_1 \\ \frac{1}{\varepsilon} 1 \end{pmatrix}$$

$$= \begin{pmatrix} \eta_1 \\ \frac{1}{\varepsilon} 1 \end{pmatrix} + \begin{pmatrix} \frac{1}{\alpha - \eta_1} & \frac{1}{\alpha - \eta_1} \vec{b}^T \\ \vec{b} \frac{1}{\alpha - \eta_1} & -\frac{1}{\varepsilon} 1 + \frac{1}{\alpha - \eta_1} \vec{b} \vec{b}^T - c \end{pmatrix} \xrightarrow{\varepsilon \rightarrow 0} \begin{pmatrix} \eta_1 + \frac{1}{\alpha - \eta_1} & \frac{1}{\alpha - \eta_1} \vec{b}^T \\ \vec{b} \frac{1}{\alpha - \eta_1} & \frac{1}{\alpha - \eta_1} \vec{b} \vec{b}^T - c \end{pmatrix}$$

We can simplify further. Consider the full Choi Covariance

$$\begin{pmatrix} \Sigma_{LL} - \Sigma_{LR} (\Sigma_{RR} - K_{LL}^{-1})^{-1} \Sigma_{LR}^T & -\Sigma_{LR} (\Sigma_{RR} - K_{LL}^{-1})^{-1} K_{LL}^{-1} K_{LR} \\ -K_{LR}^T K_{LL}^{-1} (\Sigma_{RR} - K_{LL}^{-1})^{-1} \Sigma_{LR}^T & K_{RR} - K_{LR}^T (K_{LL}^{-1} + K_{LL}^{-1} (\Sigma_{RR} - K_{LL}^{-1})^{-1} K_{LL}^{-1}) K_{LR} \end{pmatrix}$$

$$1) \Sigma_{LL} - \Sigma_{LR} (\Sigma_{RR} - K_{LL}^{-1})^{-1} \Sigma_{LR}^T = \Sigma_{LL} - \frac{1}{\alpha - \eta_1} \Sigma_{LR} P \Sigma_{LR}^T$$

$$2) -\Sigma_{LR} (\Sigma_{RR} - K_{LL}^{-1})^{-1} K_{LL}^{-1} K_{LR} = -\Sigma_{LR} \begin{pmatrix} \frac{\eta_1}{\alpha - \eta_1} & \frac{1}{\alpha - \eta_1} \vec{b}^T \\ 0 & -1 \end{pmatrix} \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{=1-P}$$

$$= \Sigma_{LR} \begin{pmatrix} 0 & -\frac{1}{\alpha - \eta_1} \vec{b}^T \\ 0 & 1 \end{pmatrix}$$

$$3) -K_{LR}^+ K_{LL}^{-1} (\Sigma_{RR} - K_{LL}^{-1})^{-1} \Sigma_{LR}^+ = \begin{pmatrix} 0 & 0 \\ \frac{-1}{a-\eta_1} \vec{b} & 1 \end{pmatrix} \Sigma_{LR}^+$$

$$4) K_{RR} - K_{LR}^+ (K_{LL}^{-1} + K_{LL}^{-1} (\Sigma_{RR} - K_{LL}^{-1})^{-1} K_{LL}^{-1}) K_{LR}$$

$$= \eta_2 P - \begin{pmatrix} 0 & \\ & 1 \end{pmatrix} \begin{pmatrix} \eta_1 + \frac{1}{a-\eta_1} & \frac{1}{a-\eta_1} \vec{b}^+ \\ \vec{b} \frac{1}{a-\eta_1} & \frac{1}{a-\eta_1} \vec{b} \vec{b}^+ - c \end{pmatrix} \begin{pmatrix} 0 & \\ & 1 \end{pmatrix}$$

$$= \eta_2 P - \left(\frac{1}{a-\eta_1} \vec{b} \vec{b}^+ - c \right) (I - P)$$

Thus, $\begin{array}{c} L \\ \hline \textcircled{\Sigma_{LL}} \\ \hline R \end{array} \xrightarrow{\varepsilon \rightarrow 0} \begin{pmatrix} \Sigma_{LL} - \frac{1}{a-\eta_1} \Sigma_{LR} P \Sigma_{LR}^+ & \Sigma_{LR} \begin{pmatrix} 0 & \frac{-1}{a-\eta_1} \vec{b}^+ \\ 0 & 1 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \\ \frac{-1}{a-\eta_1} \vec{b} & 1 \end{pmatrix} \Sigma_{LR}^+ & \eta_2 P - \left(\frac{1}{a-\eta_1} \vec{b} \vec{b}^+ - c \right) (I - P) \end{pmatrix}$

Where $\begin{cases} a \equiv (\Sigma_{RR})_{11} \\ (\vec{b})_i \equiv (\Sigma_{RR})_{i1}, i=1, \dots \\ C_{ij} \equiv (\Sigma_{RR})_{ij}, i, j \neq 1 \end{cases}$

in the basis where P is diagonal

Check: Σ .

$$, a=1, \vec{b}=0, c=1$$

$$[\eta_2 P + (1-P_2)]^2 = P^2 + (1-P_2)^2 = 1 \checkmark$$

$$\Sigma_{LL} = \Sigma_{RR} = 1,$$